

XIAO LIANG

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Education

University of California San Diego

MS & PhD in Electrical and Computer Engineering

Sep. 2021 – present

La Jolla, California

University of Washington Seattle

BS in Computer Science and Engineering

Sep. 2017 – Jun. 2021

Seattle, Washington

Industrial Experience

Applied Scientist Co-op

Amazon Robotics, Manipulation Research

Sep. 2025 – Mar. 2026

Boston, Massachusetts

- Designed and benchmarked conditioning architectures for injecting object-level physical properties into pretrained video diffusion transformers.
- Built a simulation-driven data generation pipeline with domain randomization; showed successfully transfer of physics prior from simulation to reality.
- Validated on production warehouse data from Amazon's Sparrow robot, improving mispick and swing prediction accuracy and enabling counterfactual reasoning over object mass and friction.

Selected Publications

Xiao Liang, Lu Shen, Peihan Zhang, Soofiyan Atar, Florian Richter, Michael C. Yip. Towards Autonomous Tape Handling for Robotic Wound Redressing. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

Zekai Liang, Xiao Liang, Soofiyan Atar, Sreyan Das, Zoe Chiu, Peihan Zhang, Calvin Joyce, Florian Richter, Shanglei Liu, Michael C. Yip. LapSurgie: Humanoid Robots Performing Surgery via Teleoperated Handheld Laparoscopy. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

Soofiyan Atar, Xiao Liang*, Calvin Joyce*, Florian Richter, Ricardo Wood, Charles Goldberg, Preetham Suresh, Michael Yip. Humanoids in Hospitals: A Technical Study of Humanoid Surrogates for Dexterous Medical Interventions. *arXiv preprint arXiv:2503.12725*, 2025.

Xiao Liang*, Chung-pang Wang*, Nikhil Shinde, Fei Liu, Florian Richter, Michael C. Yip. MEDiC: Autonomous Surgical Robotic Assistance to Maximizing Exposure for Dissection and Cautery. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

Xiao Liang*, Youcheng Zhang*, Fei Liu, Florian Richter, Michael C. Yip. AutoPeel: Adhesion-aware Safe Peeling Trajectory Optimization for Robotic Wound Care. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

Xiao Liang*, Nikhil Shinde*, Fei Liu, Yutong Zhang, Florian Richter, Sylvia Herbert, Michael C. Yip. JIGGLE: An Active Sensing Framework for Boundary Parameters Estimation in Deformable Surgical Environments *Robotics: Science and Systems (RSS)*, 2024.

Xiao Liang*, Fei Liu*, Yutong Zhang, Yuelei Li, Shan Lin, Michael C. Yip. Real-to-Sim Deformable Object Manipulation: Optimizing Physics Models with Residual Mappings for Robotic Surgery. *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

Xiao Liang, Shan Lin, Fei Liu, Dimitri Schreiber, Michael C. Yip. ORRN: An ODE-based Recursive Registration Network for Deformable Respiratory Motion Estimation with Lung 4DCT Images. *IEEE Transactions on Biomedical Engineering (TBME)*, 2023

Research Experience

Graduate Student Research

Advised by Professor Michael Yip, UCSD Advanced Robotics and Controls Lab

Sep 2021 – present

La Jolla, California

- Developed high-fidelity simulations of surgical environments for robot policy learning and model predictive control.

- Built deformable object manipulation algorithms with differential simulation, and incorporated feedback from robot perception and probabilistic estimation.
- Worked in depth with multiple robotic system and teleoperation platform including dVRK, Panda robot, Taurus surgical robot, and humanoid teleoperation.

Undergraduate Research Assistant

Jan 2021 – Jun 2021

Computer Vision, UW Graphics and Imaging Laboratory

Seattle, Washington

- Extending a previous research Background Matting V2, developed a novel video matting neural network that generates high-resolution matte and estimates a static background in real time.
- Improved the matte prediction by utilizing motions and developed criteria for selecting key frames for background reconstruction in a video.

Undergraduate Research Assistant

March 2020 – Jun 2021

Human Computer Interaction (Mixed Reality), UW Reality Lab

Seattle, Washington

- Developed a deep learning-enabled Mixed Reality application using Unity for augmenting the cooking experience.
- Trained and deployed SOTA object recognition and detection neural networks on a cloud server.
- Developed near real-time communication scheme between an Magic Leap One headset and the cloud machine, providing semantic awareness to the mixed reality device.

Teaching Experience

Teaching Assistant

Mar. 2021 – Jun. 2021

CSE 481V Virtual Reality Capstone, University of Washington

Seattle, Washington

Teaching Assistant

Jun. 2020 – Dec. 2020

CSE 457 Computer Graphics, University of Washington

Seattle, Washington

Other Projects

Neural Process for Safe Exploration | *Neural Process*

- Developed a neural process guided safe exploration algorithm for a movie recommendation problem.

Volume Rendering in Virtual Reality System | *Unity, High Level Shading Language*

- Developed a real-time, interactive, volume rendering algorithm for visualizing 3D medical image in Virtual Reality.

Graph Neural Network Particle Simulator | *Graph Neural Network*

- Implemented a previous work on using Graph neural network for simulating fluid particle dynamics.

Human Pose-controlled Game | *Kinect, Unity, Neural Network*

- Made a Super Mario-like game that is controlled by player's poses classified by a neural network using Kinect's data.

Technical Skills

Languages: Python, Java, C++, Matlab, JavaScript

Tools & libraries: Nvidia Warp, Curobo, Omniverse, Pytorch, Jax, Numpy, OMPL, Unity, Blender, Slicer